2024-25 First Semester

Calculus 1

Assignment 5

Due Date: 25/July/2024(Mon.), on or before 17:00

- Write down your CHN name and student ID. Write neatly on A4-sized paper (staple if necessary) and show your steps.
- For online students, hand in your homework in one pdf file on iSpace.
- Late submissions or answers without details will not be graded.

---

1. Calculate y'.

(1) $y=\left(x^2+x^3\right)^4$

(2) $y=\frac{x^2-x+2}{\sqrt{x}}$

(3) $y=x^2\sin\pi x$

(4) $y=\ln(x\ln x)$

(5) $y=\sqrt{x}\cos\sqrt{x}$

(6) $y=\frac{e^{1/x}}{x^2}$

(7) $y=\tan\left(\frac{t}{1+t^2}\right)$

(8) $y=3^{x\ln x}$

(9) $y=\left(1-x^{-1}\right)^{-1}$

(10) $y=\log_5(1+2x)$

(11) $y=\cos\left(e^{\sqrt{\tan 3x}}\right)$

(12) $y=\sin^2(\cos\sqrt{\sin\pi x})$

Sol:

(1) $y=\left(x^2+x^3\right)^4\Rightarrow$

$$y'=4\left(x^2+x^3\right)^3\left(2x+3x^2\right)=4\left(x^2\right)^3(1+x)^3x(2+3x)=4x^7(x+1)^3(3x+2)$$

(2) $y=\frac{x^2-x+2}{\sqrt{x}}=x^{3/2}-x^{1/2}+2x^{-1/2}\Rightarrow y'=\frac{3}{2}x^{1/2}-\frac{1}{2}x^{-1/2}-x^{-3/2}=\frac{3}{2}\sqrt{x}-\frac{1}{2\sqrt{x}}-\frac{1}{\sqrt{x^3}}$

(3) $y=x^2\sin\pi x \quad\Rightarrow\quad y'=x^2(\cos\pi x)\pi+(\sin\pi x)(2x)=x(\pi x\cos\pi x+2\sin\pi x)$

(4)

$$y=\ln(x\ln x)\Rightarrow y'=\frac{1}{x\ln x}(x\ln x)'=\frac{1}{x\ln x}\left(x\cdot\frac{1}{x}+\ln x\cdot 1\right)=\frac{1+\ln x}{x\ln x}$$

Another method: $y=\ln(x\ln x)=\ln x+\ln\ln x\Rightarrow y'=\frac{1}{x}+\frac{1}{\ln x}\cdot\frac{1}{x}=\frac{\ln x+1}{x\ln x}$ **打开**ln分别求导

$y=\sqrt{x}\cos\sqrt{x}\Rightarrow$

(5) $y'=\sqrt{x}(\cos\sqrt{x})'+\cos\sqrt{x}(\sqrt{x})'=\sqrt{x}\left[-\sin\sqrt{x}\left(\frac{1}{2}x^{-1/2}\right)\right]+\cos\sqrt{x}\left(\frac{1}{2}x^{-1/2}\right)$

$$=\frac{1}{2}x^{-1/2}(-\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x})=\frac{\cos\sqrt{x}-\sqrt{x}\sin\sqrt{x}}{2\sqrt{x}}$$

(6) $y=\frac{e^{1/x}}{x^2}\Rightarrow y'=\frac{x^2\left(e^{1/x}\right)'-e^{1/x}\left(x^2\right)'}{\left(x^2\right)^2}=\frac{x^2\left(e^{1/x}\right)\left(-1/x^2\right)-e^{1/x}(2x)}{x^4}=\frac{-e^{1/x}(1+2x)}{x^4}$

(7) $y=\tan\left(\frac{t}{1+t^2}\right)\Rightarrow$

$$y'=\sec^2\left(\frac{t}{1+t^2}\right)\frac{d}{dt}\left(\frac{t}{1+t^2}\right)=\sec^2\left(\frac{t}{1+t^2}\right)\cdot\frac{\left(1+t^2\right)(1)-t(2t)}{\left(1+t^2\right)^2}=\frac{1-t^2}{\left(1+t^2\right)^2}\sec^2\left(\frac{t}{1+t^2}\right)$$

(8) $y=3^{x\ln x}\Rightarrow y'=3^{x\ln x}(\ln 3)\frac{d}{dx}(x\ln x)=3^{x\ln x}(\ln 3)\left(x\cdot\frac{1}{x}+\ln x\cdot 1\right)=3^{x\ln x}(\ln 3)(1+\ln x)$

(9) $y=\left(1-x^{-1}\right)^{-1}\Rightarrow$

$$y'=-1\left(1-x^{-1}\right)^{-2}\left[-\left(-1x^{-2}\right)\right]=-(1-1/x)^{-2}x^{-2}=-((x-1)/x)^{-2}x^{-2}=-(x-1)^{-2}$$

(10) $y=\log_5(1+2x)\Rightarrow y'=\frac{1}{(1+2x)\ln 5}\frac{d}{dx}(1+2x)=\frac{2}{(1+2x)\ln 5}$

(11) $y=\cos\left(e^{\sqrt{\tan 3x}}\right)\Rightarrow$

$$y'=-\sin\left(e^{\sqrt{\tan 3x}}\right)\cdot\left(e^{\sqrt{\tan 3x}}\right)'=-\sin\left(e^{\sqrt{\tan 3x}}\right)e^{\sqrt{\tan 3x}}\cdot\frac{1}{2}(\tan 3x)^{-1/2}\cdot\sec^2(3x)\cdot 3$$
$$=\frac{-3\sin\left(e^{\sqrt{\tan 3x}}\right)e^{\sqrt{\tan 3x}}\sec^2(3x)}{2\sqrt{\tan 3x}}$$

(12) $y=\sin^2(\cos\sqrt{\sin\pi x})=[\sin(\cos\sqrt{\sin\pi x})]^2\Rightarrow$

$$y'=2[\sin(\cos\sqrt{\sin\pi x})][\sin(\cos\sqrt{\sin\pi x})]'=2\sin(\cos\sqrt{\sin\pi x})\cos(\cos\sqrt{\sin\pi x})(\cos\sqrt{\sin\pi x})'$$
$$=2\sin(\cos\sqrt{\sin\pi x})\cos(\cos\sqrt{\sin\pi x})(-\sin\sqrt{\sin\pi x})(\sqrt{\sin\pi x})'$$
$$=-2\sin(\cos\sqrt{\sin\pi x})\cos(\cos\sqrt{\sin\pi x})\sin\sqrt{\sin\pi x}\cdot\frac{1}{2}(\sin\pi x)^{-1/2}(\sin\pi x)'$$
$$=\frac{-\sin(\cos\sqrt{\sin\pi x})\cos(\cos\sqrt{\sin\pi x})\sin\sqrt{\sin\pi x}}{\sqrt{\sin\pi x}}\cdot\cos\pi x\cdot\pi$$
$$=\frac{-\pi\sin(\cos\sqrt{\sin\pi x})\cos(\cos\sqrt{\sin\pi x})\sin\sqrt{\sin\pi x}\cos\pi x}{\sqrt{\sin\pi x}}$$

2. If $g(x)=x/e^x$, find $g^{(n)}(x)$.
Sol:

$$g(x)=\frac{x}{e^x}\Rightarrow g'(x)=\frac{e^x\cdot 1-x\cdot e^x}{(e^x)^2}=\frac{e^x(1-x)}{(e^x)^2}=\frac{1-x}{e^x}\Rightarrow$$
$$g''(x)=\frac{e^x\cdot(-1)-(1-x)e^x}{(e^x)^2}=\frac{e^x[-1-(1-x)]}{(e^x)^2}=\frac{x-2}{e^x}\Rightarrow$$
$$g'''(x)=\frac{e^x\cdot 1-(x-2)e^x}{(e^x)^2}=\frac{e^x[1-(x-2)]}{(e^x)^2}=\frac{3-x}{e^x}\Rightarrow$$
$$g^{(4)}(x)=\frac{e^x\cdot(-1)-(3-x)e^x}{(e^x)^2}=\frac{e^x[-1-(3-x)]}{(e^x)^2}=\frac{x-4}{e^x}.$$

The pattern suggests that $g^{(n)}(x)=\frac{(x-n)(-1)^n}{e^x}$. (We could use mathematical induction to prove this formula.)

3. Use the Reciprocal Rule and Product Rule to prove the Quotient Rule.
Sol:
Reciprocal rule $\left(\frac{1}{g}\right)' = \frac{-g'}{g^2}$
product rule $(fg)' = f'g + fg'$
Quotient rule $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

$$\begin{aligned}\left(\frac{f}{g}\right)' &= \left(f \cdot \frac{1}{g}\right)' \\ &= f'\frac{1}{g} + f\left(\frac{1}{g}\right)' \\ &= f'\frac{1}{g} + f\left(\frac{-g'}{g^2}\right) \\ &= \frac{f'g - fg'}{g^2}\end{aligned}$$

4. (a) Evaluate $\lim_{x\to\infty} x\sin\frac{1}{x}$. (b) Evaluate $\lim_{x\to 0} x\sin\frac{1}{x}$.
Sol:
(a) Let $\theta = \frac{1}{x}$. Then as $x \to \infty, \theta \to 0^+$, and $\lim_{x\to\infty} x\sin\frac{1}{x} = \lim_{\theta\to 0+}\frac{1}{\theta}\sin\theta = \lim_{\theta\to 0}\frac{\sin\theta}{\theta} = 1$.
(b) Since $-1 \le \sin(1/x) \le 1$, we have (as illustrated in the figure) $-|x| \le x\sin(1/x) \le |x|$. We know that $\lim_{x\to 0}(|x|) = 0$ and $\lim_{x\to 0}(-|x|) = 0$; so by the Squeeze Theorem, $\lim_{x\to 0} x\sin(1/x) = 0$.

5. Find the limit.
(1) $\lim_{x\to 0}\frac{\sin 5x}{3x}$
(2)$\lim_{t\to 0}\frac{\tan 6t}{\sin 2t}$
(3)$\lim_{x\to 0}\frac{\sin 3x}{5x^3-4x}$
(4)$\lim_{\theta\to 0}\frac{\sin\theta}{\theta+\tan\theta}$
(5) $\lim_{x\to 0}\csc x\sin(\sin x)$
(6) $\lim_{x\to 1}\frac{\sin(x-1)}{x^2+x-2}$
Sol:
(1)$\lim_{x\to 0}\frac{\sin 5x}{3x} = \lim_{x\to 0}\frac{5}{3}\left(\frac{\sin 5x}{5x}\right) = \frac{5}{3}\lim_{x\to 0}\frac{\sin 5x}{5x} = \frac{5}{3}\lim_{\theta\to 0}\frac{\sin\theta}{\theta}$ $[\theta = 5x]$ $= \frac{5}{3}\cdot 1 = \frac{5}{3}$
(2)

$$\begin{aligned}\lim_{t\to 0}\frac{\tan 6t}{\sin 2t} &= \lim_{t\to 0}\left(\frac{\sin 6t}{t}\cdot\frac{1}{\cos 6t}\cdot\frac{t}{\sin 2t}\right) = \lim_{t\to 0}\frac{6\sin 6t}{6t}\cdot\lim_{t\to 0}\frac{1}{\cos 6t}\cdot\lim_{t\to 0}\frac{2t}{2\sin 2t} \\ &= 6\lim_{t\to 0}\frac{\sin 6t}{6t}\cdot\lim_{t\to 0}\frac{1}{\cos 6t}\cdot\frac{1}{2}\lim_{t\to 0}\frac{2t}{\sin 2t} = 6(1)\cdot\frac{1}{1}\cdot\frac{1}{2}(1) = 3\end{aligned}$$

(3)$\lim_{x\to 0}\frac{\sin 3x}{5x^3-4x} = \lim_{x\to 0}\left(\frac{\sin 3x}{3x}\cdot\frac{3}{5x^2-4}\right) = \lim_{x\to 0}\frac{\sin 3x}{3x}\cdot\lim_{x\to 0}\frac{3}{5x^2-4} = 1\cdot\left(\frac{3}{-4}\right) = -\frac{3}{4}$
(4) Divide numerator and denominator by θ. ($\sin\theta$ also works.)

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta+\tan\theta} = \lim_{\theta\to 0}\frac{\frac{\sin\theta}{\theta}}{1+\frac{\sin\theta}{\theta}\cdot\frac{1}{\cos\theta}} = \frac{\lim_{\theta\to 0}\frac{\sin\theta}{\theta}}{1+\lim_{\theta\to 0}\frac{\sin\theta}{\theta}\lim_{\theta\to 0}\frac{1}{\cos\theta}} = \frac{1}{1+1\cdot 1} = \frac{1}{2}$$

(5) $\lim_{x\to 0}\csc x\sin(\sin x)=\lim_{x\to 0}\frac{\sin(\sin x)}{\sin x}=\lim_{\theta\to 0}\frac{\sin\theta}{\theta}$ [As $x\to 0,\theta=\sin x\to 0$] =.
(6)$\lim_{x\to 1}\frac{\sin(x-1)}{x^2+x-2}=\lim_{x\to 1}\frac{\sin(x-1)}{(x+2)(x-1)}=\lim_{x\to 1}\frac{1}{x+2}\lim_{x\to 1}\frac{\sin(x-1)}{x-1}=\frac{1}{3}\cdot 1=\frac{1}{3}$

6. Suppose f is differentiable on $\mathbb{R}$. Let $F(x)=f(e^x)$ and $G(x)=e^{f(x)}$. Find expressions for (a) $F'(x)$ and (b) $G'(x)$.
Sol:
(a) $F(x)=f(e^x)\Rightarrow F'(x)=f'(e^x)\frac{d}{dx}(e^x)=f'(e^x)e^x$
(b) $G(x)=e^{f(x)}\Rightarrow G'(x)=e^{f(x)}\frac{d}{dx}f(x)=e^{f(x)}f'(x)$

7. If $y=f(u)$ and $u=g(x)$, where f and g are twice differentiable functions, show that

$$\frac{d^2y}{dx^2}=\frac{d^2y}{du^2}\left(\frac{du}{dx}\right)^2+\frac{dy}{du}\frac{d^2u}{dx^2}$$

Sol:
The Chain Rule says that $\frac{dy}{dx}=\frac{dy}{du}\frac{du}{dx}$, so

$$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)=\frac{d}{dx}\left(\frac{dy}{du}\frac{du}{dx}\right)=\left[\frac{d}{dx}\left(\frac{dy}{du}\right)\right]\frac{du}{dx}+\frac{dy}{du}\frac{d}{dx}\left(\frac{du}{dx}\right)\quad\text{[Product Rule]}$$
$$=\left[\frac{d}{du}\left(\frac{dy}{du}\right)\frac{du}{dx}\right]\frac{du}{dx}+\frac{dy}{du}\frac{d^2u}{dx^2}=\frac{d^2y}{du^2}\left(\frac{du}{dx}\right)^2+\frac{dy}{du}\frac{d^2u}{dx^2}$$

8. Find the linearization $L(x)$ of the function at a.
$f(x)=\sqrt{x},\quad a=4$
Sol:
$f(x)=\sqrt{x}\Rightarrow f'(x)=\frac{1}{2}x^{-1/2}=1/(2\sqrt{x})$, so $f(4)=2$ and $f'(4)=\frac{1}{4}$. Thus, $L(x)=f(4)+f'(4)(x-4)=2+\frac{1}{4}(x-4)=2+\frac{1}{4}x-1=\frac{1}{4}x+1$.

9. Use a linear approximation (or differentials) to estimate the given number.
$e^{0.1}$ Sol:
$y=f(x)=e^x\Rightarrow dy=e^x dx$. When $x=0$ and $dx=0.1, dy=e^0(0.1)=0.1$, so $e^{0.1}=f(0.1)\approx f(0)+dy=1+0.1=1.1$.

Bonus:
1. Evaluate $\lim_{x\to\pi}\frac{e^{\sin x}-1}{x-\pi}$.
Sol:
We recognize this limit as the definition of the derivative of the function $f(x)=e^{\sin x}$ at $x=\pi$, since it is of the form $\lim_{x\to\pi}\frac{f(x)-f(\pi)}{x-\pi}$. Therefore, the limit is equal to $f'(\pi)=(\cos\pi)e^{\sin\pi}=-1\cdot e^0=-1$.
2. If f and g are differentiable functions with $f(0)=g(0)=0$ and $g'(0)\neq 0$, show that

$$\lim_{x\to 0}\frac{f(x)}{g(x)}=\frac{f'(0)}{g'(0)}$$

Sol:

$\lim_{x\to 0}\frac{f(x)}{g(x)}=\lim_{x\to 0}\frac{f(x)-0}{g(x)-0}=\lim_{x\to 0}\frac{f(x)-f(0)}{g(x)-g(0)}=\lim_{x\to 0}\frac{\frac{f(x)-f(0)}{x-0}}{\frac{g(x)-g(0)}{x-0}}=\frac{\lim_{x\to 0}\frac{f(x)-f(0)}{x-0}}{\lim_{x\to 0}\frac{g(x)-g(0)}{x-0}}=\frac{f'(0)}{g'(0)}$